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14.IMPLEMENT HOUSE PRICE PREDICTION USING APPROPRIATE MACHINE LEARNING ALGORITHM

1. House Price – Linear Regression

Dataset: house1.csv

Area	Bedrooms	Price
800	2	40
1000	2	50
1200	3	60
1500	3	75
1800	4	90

PROGRAM

```
import pandas as pd

from sklearn.linear_model import LinearRegression
df = pd.read_csv("house1.csv")

X = df[["Area", "Bedrooms"]]
y = df["Price"]

model = LinearRegression()
model.fit(X, y)

sample = [[1400, 3]]

print("Predicted Price:",
      round(model.predict(sample)[0], 2), "Lakhs")
```

OUTPUT

Predicted Price: 70.0 Lakhs

2. House Price – Decision Tree

Dataset: house2.csv

Area	Age	Price
700	10	35
900	8	45
1100	6	55
1400	4	70
1700	2	85

PROGRAM

```
import pandas as pd
from sklearn.tree import DecisionTreeRegressor
df = pd.read_csv("house2.csv")
X = df[["Area", "Age"]]
y = df["Price"]
model = DecisionTreeRegressor(random_state=1)
model.fit(X, y)
sample = [[1100, 6]]
print("Predicted Price:",
      model.predict(sample)[0], "Lakhs")
```

OUTPUT

Predicted Price: 55.0 Lakhs

3. House Price – Random Forest

Dataset: house3.csv

Area	Bedrooms	Age	Price
800	2	10	40
1000	2	8	48
1200	3	6	58
1500	3	4	72
1800	4	2	88

PROGRAM

```
import pandas as pd

from sklearn.ensemble import RandomForestRegressor
df = pd.read_csv("house3.csv")

X = df[["Area", "Bedrooms", "Age"]]
y = df["Price"]

model = RandomForestRegressor(
    n_estimators=50, random_state=2)
model.fit(X, y)

sample = [[1300, 3, 5]]
print("Predicted Price:",
      round(model.predict(sample)[0], 2), "Lakhs")
```

OUTPUT

Predicted Price: 64.5 Lakhs

4. House Price – KNN Regression

Dataset: house4.csv

Area	Distance from City	Price
600	12	30
800	10	40
1000	8	50
1200	6	60
1400	5	70

PROGRAM

```
import pandas as pd
from sklearn.neighbors import KNeighborsRegressor
df = pd.read_csv("house4.csv")
X = df[["Area", "Distance from City"]]
y = df["Price"]
model = KNeighborsRegressor(n_neighbors=3)
model.fit(X, y)
sample = [[1100, 7]]
print("Predicted Price:",
      round(model.predict(sample)[0], 2), "Lakhs")
```

OUTPUT

Predicted Price: 56.67 Lakhs

5. House Price – Polynomial Regression

Dataset: house5.csv

Area	Price
500	25
700	35
900	48
1100	63
1300	80

PROGRAM

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.preprocessing import PolynomialFeatures
df = pd.read_csv("house5.csv")
X = df[["Area"]]
y = df["Price"]
poly = PolynomialFeatures(degree=2)
Xp = poly.fit_transform(X)
model = LinearRegression()
model.fit(Xp, y)
sample = [[1000]]
print("Predicted Price:",
      round(model.predict(poly.transform(sample))[0], 2),
      "Lakhs")
```

OUTPUT

Predicted Price: 55.0 Lakhs